

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		Mass	1			1		
		(ii)		Body at rest remains at rest, body in motion remains at {constant velocity / constant speed in a straight line} (1) unless {resultant / external / unbalanced} force acts (1) OWTTE	2			2		
	(b)	(i)		[Applied] force - accept {mass / weight} of the hanger	1			1		1
		(ii)		Acceleration	1			1		1
		(iii)		Total mass or surface or the distance travelled	1			1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		When force doubles [from 0.5 to 1.0] (1) acceleration [roughly] doubles from 0.310 to 0.619 (1) so agree Conclusion must be present to award both marks Or When force triples [from 0.5 to 1.5] (1) acceleration [roughly] triples from 0.310 to 0.931 (1) so agree Conclusion must be present to award both marks Or using ratios: 1 mark for 1 ratio correctly calculated 2 marks for 2 ratios correctly calculated with so agree i.e. $\frac{0.310}{0.5} = 0.62$; $\frac{0.619}{1.0} = 0.62$; $\frac{0.931}{1.5} = 0.62$ or reciprocals Conclusion must be present to award both marks Or An increase of 0.5 [N] (1) Produces an increased acceleration of approximately 0.310 [m/s²] each time (1) Conclusion must be present to award both marks			2	2	2	2
		(v)	I	$0.5 = m \times 0.310$ (1) $m = \frac{0.5}{0.310}$ (1) $m = 1.61$ or 1.6 [kg] (1)	1	1 1		3	3	3
			II	Use a balance to determine the mass of the trolley and the slotted masses		1		1		1
				Question 8 total	7	3	2	12	5	9